

Comparing TurboQuant+ with traditional quantization methods on real tasks

Stanford CS224N {Custom, Default} Project

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Abstract

An abstract should concisely (less than 300 words) motivate the problem, describe your aims, describe your contribution, and highlight your main finding(s).

1 Key Information to include

- Mentor:
- External Collaborators (if you have any):
- Sharing project:

2 Introduction

The introduction explains the problem, why it's difficult, interesting, or important, how and why current methods succeed/fail at the problem, and explains the key ideas of your approach and results. Though an introduction covers similar material as an abstract, the introduction gives more space for motivation, detail, references to existing work, and to capture the reader's interest.

3 Related Work

This section helps the reader understand the research context of your work by providing an overview of existing work in the area.

4 Approach

This section details your approach(es) to the problem. For example, this is where you describe the architecture of your neural network(s), and any other key methods or algorithms.

5 Experiments

This section contains the following.

5.1 Data

Describe the dataset(s) you are using (provide references). If it's not already clear, make sure the associated task is clearly described. Being precise about the exact form of the input and output can be very useful for readers attempting to understand your work, especially if you've defined your own task.

5.2 Evaluation method

Describe the evaluation metric(s) you use, plus any other details necessary to understand your evaluation. Some projects will have clear metrics from prior work on given datasets, but we realize that other projects will define their own metrics. If you're defining your own metrics, be clear as to what you're hoping to measure with each evaluation method (whether quantitative or qualitative, automatic or human-defined!), and how it's defined.

5.3 Experimental details

Report how you ran your experiments (e.g., model configurations, learning rate, training time, etc.)

5.4 Results

Report the quantitative results that you have found. Use a table or plot to compare results and compare against baselines.

- If you're a default project team, you should **report the accuracy and Pearson correlation scores you obtained on the test leaderboard** in this section. You can also report dev set results if you'd like.
- Comment on your quantitative results. Are they what you expected? Better than you expected? Worse than you expected? Why do you think that is? What does that tell you about your approach?

6 Analysis

Your report should include *qualitative evaluation*. That is, try to understand your system (e.g., how it works, when it succeeds and when it fails) by inspecting key characteristics or outputs of your model.

7 Conclusion

Summarize the main findings of your project and what you have learned. Highlight your achievements, and note the primary limitations of your work. If you'd like, you can describe avenues for future work.

8 Ethics Statement

What are the ethical challenges and possible societal risks of your project, and what are mitigation strategies?

References

A Appendix (optional)

If you wish, you can include an appendix, which should be part of the main PDF, and does not count towards the 6-8 page limit. Appendices can be useful to supply extra details, examples, figures, results, visualizations, etc. that you couldn't fit into the main paper. However, your grader *does not* have to read your appendix, and you should assume that you will be graded based on the content of the main part of your paper only.